Wine Quality Project

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Data Exploration

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(ggplot2)

wine <- read.csv("C:/Users/est314/Downloads/winequality-red.csv", header=TRUE
, sep=";")
View(wine)
head(wine)

##   fixed.acidity volatile.acidity citric.acid residual.sugar chlorides
## 1           7.4             0.70        0.00            1.9     0.076
## 2           7.8             0.88        0.00            2.6     0.098
## 3           7.8             0.76        0.04            2.3     0.092
## 4          11.2             0.28        0.56            1.9     0.075
## 5           7.4             0.70        0.00            1.9     0.076
## 6           7.4             0.66        0.00            1.8     0.075
##   free.sulfur.dioxide total.sulfur.dioxide density   pH sulphates alcohol
## 1                  11                   34  0.9978 3.51      0.56     9.4
## 2                  25                   67  0.9968 3.20      0.68     9.8
## 3                  15                   54  0.9970 3.26      0.65     9.8
## 4                  17                   60  0.9980 3.16      0.58     9.8
## 5                  11                   34  0.9978 3.51      0.56     9.4
## 6                  13                   40  0.9978 3.51      0.56     9.4
##   quality
## 1       5
## 2       5
## 3       5
## 4       6
## 5       5
## 6       5
```

```r
summary(wine)
```

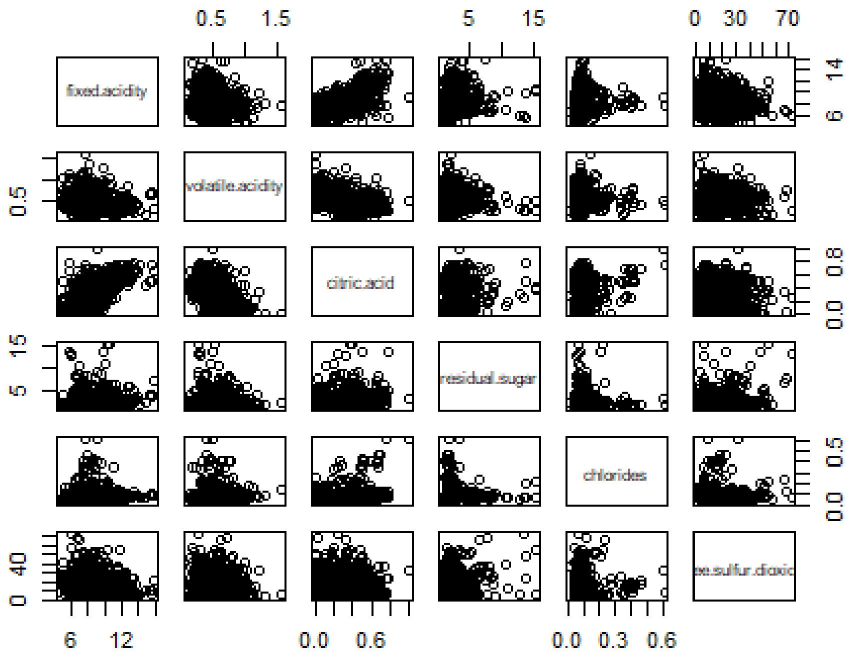
```
##  fixed.acidity   volatile.acidity  citric.acid    residual.sugar  
##  Min.   : 4.60   Min.   :0.1200   Min.   :0.000   Min.   : 0.900  
##  1st Qu.: 7.10   1st Qu.:0.3900   1st Qu.:0.090   1st Qu.: 1.900  
##  Median : 7.90   Median :0.5200   Median :0.260   Median : 2.200  
##  Mean   : 8.32   Mean   :0.5278   Mean   :0.271   Mean   : 2.539  
##  3rd Qu.: 9.20   3rd Qu.:0.6400   3rd Qu.:0.420   3rd Qu.: 2.600  
##  Max.   :15.90   Max.   :1.5800   Max.   :1.000   Max.   :15.500  
##    chlorides       free.sulfur.dioxide total.sulfur.dioxide    density      
##  Min.   :0.01200   Min.   : 1.00       Min.   :  6.00       Min.   :0.9901  
##  1st Qu.:0.07000   1st Qu.: 7.00       1st Qu.: 22.00       1st Qu.:0.9956  
##  Median :0.07900   Median :14.00       Median : 38.00       Median :0.9968  
##  Mean   :0.08747   Mean   :15.87       Mean   : 46.47       Mean   :0.9967  
##  3rd Qu.:0.09000   3rd Qu.:21.00       3rd Qu.: 62.00       3rd Qu.:0.9978  
##  Max.   :0.61100   Max.   :72.00       Max.   :289.00       Max.   :1.0037  
##        pH          sulphates         alcohol         quality     
##  Min.   :2.740   Min.   :0.3300   Min.   : 8.40   Min.   :3.000  
##  1st Qu.:3.210   1st Qu.:0.5500   1st Qu.: 9.50   1st Qu.:5.000  
##  Median :3.310   Median :0.6200   Median :10.20   Median :6.000  
##  Mean   :3.311   Mean   :0.6581   Mean   :10.42   Mean   :5.636  
##  3rd Qu.:3.400   3rd Qu.:0.7300   3rd Qu.:11.10   3rd Qu.:6.000  
##  Max.   :4.010   Max.   :2.0000   Max.   :14.90   Max.   :8.000
```

```r
glimpse(wine)
```

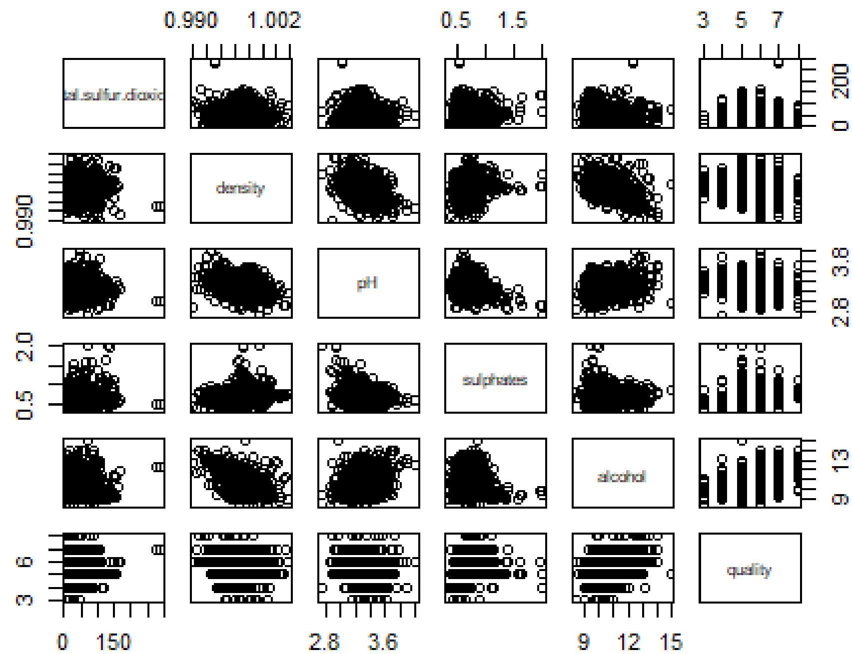
```
## Rows: 1,599
## Columns: 12
## $ fixed.acidity        <dbl> 7.4, 7.8, 7.8, 11.2, 7.4, 7.4, 7.9, 7.3, 7.8, 
7.5…
## $ volatile.acidity     <dbl> 0.700, 0.880, 0.760, 0.280, 0.700, 0.660, 0.6
00, …
## $ citric.acid          <dbl> 0.00, 0.00, 0.04, 0.56, 0.00, 0.00, 0.06, 0.0
0, 0…
## $ residual.sugar       <dbl> 1.9, 2.6, 2.3, 1.9, 1.9, 1.8, 1.6, 1.2, 2.0, 
6.1,…
## $ chlorides            <dbl> 0.076, 0.098, 0.092, 0.075, 0.076, 0.075, 0.0
69, …
## $ free.sulfur.dioxide  <dbl> 11, 25, 15, 17, 11, 13, 15, 15, 9, 17, 15, 17
, 16…
## $ total.sulfur.dioxide <dbl> 34, 67, 54, 60, 34, 40, 59, 21, 18, 102, 65, 
102,…
## $ density              <dbl> 0.9978, 0.9968, 0.9970, 0.9980, 0.9978, 0.997
8, 0…
## $ pH                   <dbl> 3.51, 3.20, 3.26, 3.16, 3.51, 3.51, 3.30, 3.3
9, 3…
## $ sulphates            <dbl> 0.56, 0.68, 0.65, 0.58, 0.56, 0.56, 0.46, 0.4
7, 0…
## $ alcohol              <dbl> 9.4, 9.8, 9.8, 9.8, 9.4, 9.4, 9.4, 10.0, 9.5, 
10.…
```

```
## $ quality                    <int> 5, 5, 5, 6, 5, 5, 5, 7, 7, 5, 5, 5, 5, 5, 5,
5, 7…

pairs(wine[, 1:6])
```




```
pairs(wine[, 7:12])
```



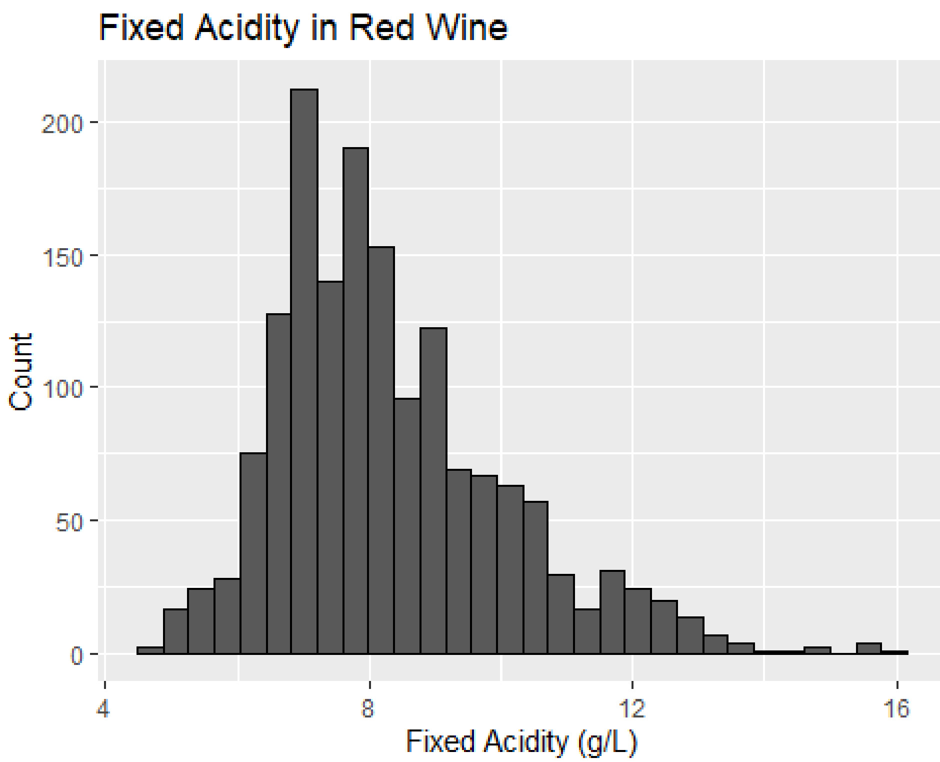

```
cor(wine)
```

```
##                      fixed.acidity volatile.acidity citric.acid residual.s
ugar
## fixed.acidity           1.00000000     -0.256130895  0.67170343    0.11477
6724
## volatile.acidity       -0.25613089      1.000000000 -0.55249568    0.00191
7882
## citric.acid             0.67170343     -0.552495685  1.00000000    0.14357
7162
## residual.sugar          0.11477672      0.001917882  0.14357716    1.00000
0000
## chlorides               0.09370519      0.061297772  0.20382291    0.05560
9535
## free.sulfur.dioxide    -0.15379419     -0.010503827 -0.06097813    0.18704
8995
## total.sulfur.dioxide   -0.11318144      0.076470005  0.03553302    0.20302
7882
## density                 0.66804729      0.022026232  0.36494718    0.35528
3371
## pH                     -0.68297819      0.234937294 -0.54190414   -0.08565
2422
## sulphates               0.18300566     -0.260986685  0.31277004    0.00552
7121
## alcohol                -0.06166827     -0.202288027  0.10990325    0.04207
5437
## quality                 0.12405165     -0.390557780  0.22637251    0.01373
```

```
1637
##                       chlorides free.sulfur.dioxide total.sulfur.dioxide
## fixed.acidity        0.093705186        -0.153794193          -0.11318144
## volatile.acidity     0.061297772        -0.010503827           0.07647000
## citric.acid          0.203822914        -0.060978129           0.03553302
## residual.sugar       0.055609535         0.187048995           0.20302788
## chlorides            1.000000000         0.005562147           0.04740047
## free.sulfur.dioxide  0.005562147         1.000000000           0.66766645
## total.sulfur.dioxide 0.047400468         0.667666450           1.00000000
## density              0.200632327        -0.021945831           0.07126948
## pH                  -0.265026131         0.070377499          -0.06649456
## sulphates            0.371260481         0.051657572           0.04294684
## alcohol             -0.221140545        -0.069408354          -0.20565394
## quality             -0.128906560        -0.050656057          -0.18510029
##                          density          pH    sulphates     alcohol
## fixed.acidity        0.66804729 -0.68297819  0.183005664 -0.06166827
## volatile.acidity     0.02202623  0.23493729 -0.260986685 -0.20228803
## citric.acid          0.36494718 -0.54190414  0.312770044  0.10990325
## residual.sugar       0.35528337 -0.08565242  0.005527121  0.04207544
## chlorides            0.20063233 -0.26502613  0.371260481 -0.22114054
## free.sulfur.dioxide -0.02194583  0.07037750  0.051657572 -0.06940835
## total.sulfur.dioxide 0.07126948 -0.06649456  0.042946836 -0.20565394
## density              1.00000000 -0.34169933  0.148506412 -0.49617977
## pH                  -0.34169933  1.00000000 -0.196647602  0.20563251
## sulphates            0.14850641 -0.19664760  1.000000000  0.09359475
## alcohol             -0.49617977  0.20563251  0.093594750  1.00000000
## quality             -0.17491923 -0.05773139  0.251397079  0.47616632
##                          quality
## fixed.acidity        0.12405165
## volatile.acidity    -0.39055778
## citric.acid          0.22637251
## residual.sugar       0.01373164
## chlorides           -0.12890656
## free.sulfur.dioxide -0.05065606
## total.sulfur.dioxide -0.18510029
## density             -0.17491923
## pH                  -0.05773139
## sulphates            0.25139708
## alcohol              0.47616632
## quality              1.00000000
```

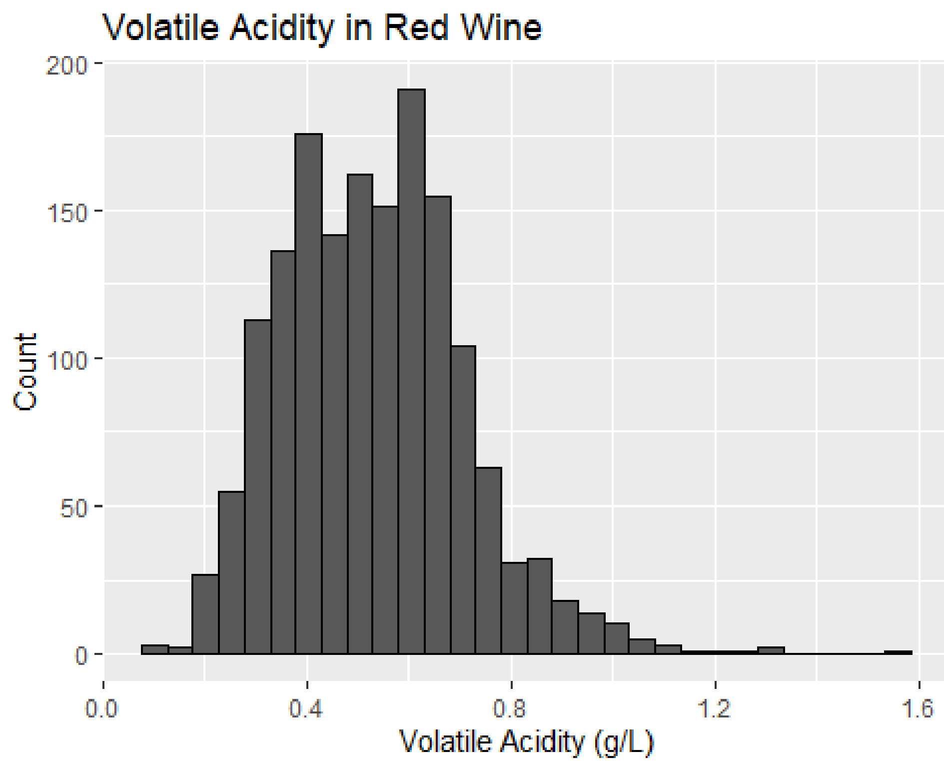
```r
ggplot(wine, aes(x = fixed.acidity)) +
  geom_histogram(color = "black") +
  labs(
    title = "Fixed Acidity in Red Wine",
    x = "Fixed Acidity (g/L)",
    y = "Count"
  )
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



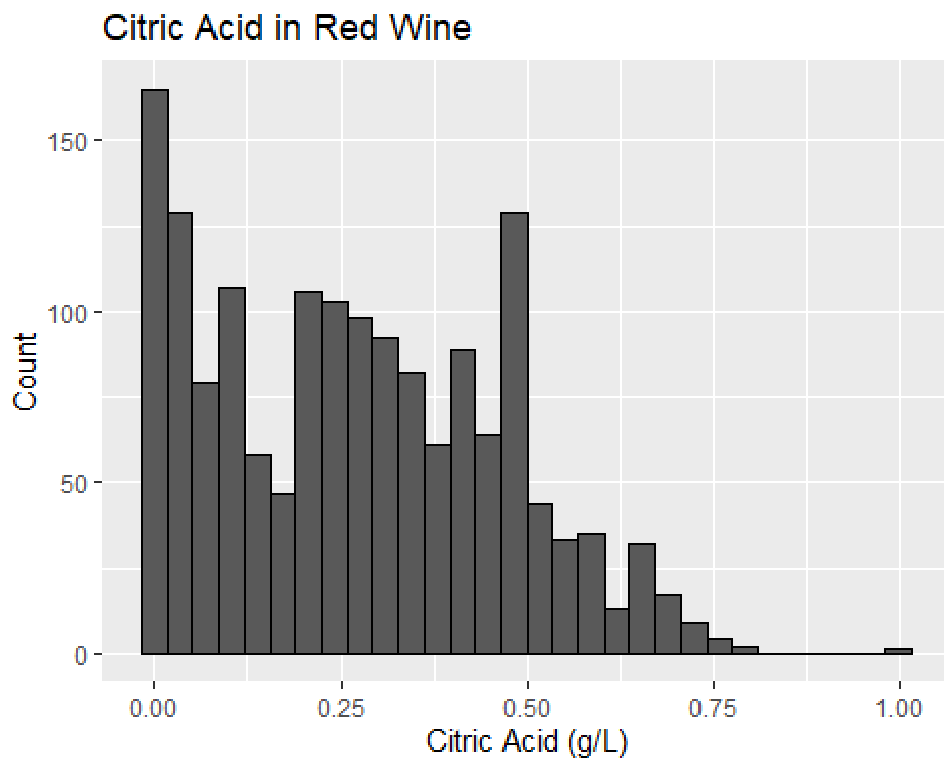

```r
ggplot(wine, aes(x = volatile.acidity)) +
  geom_histogram(color = "black") +
  labs(
    title = "Volatile Acidity in Red Wine",
    x = "Volatile Acidity (g/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



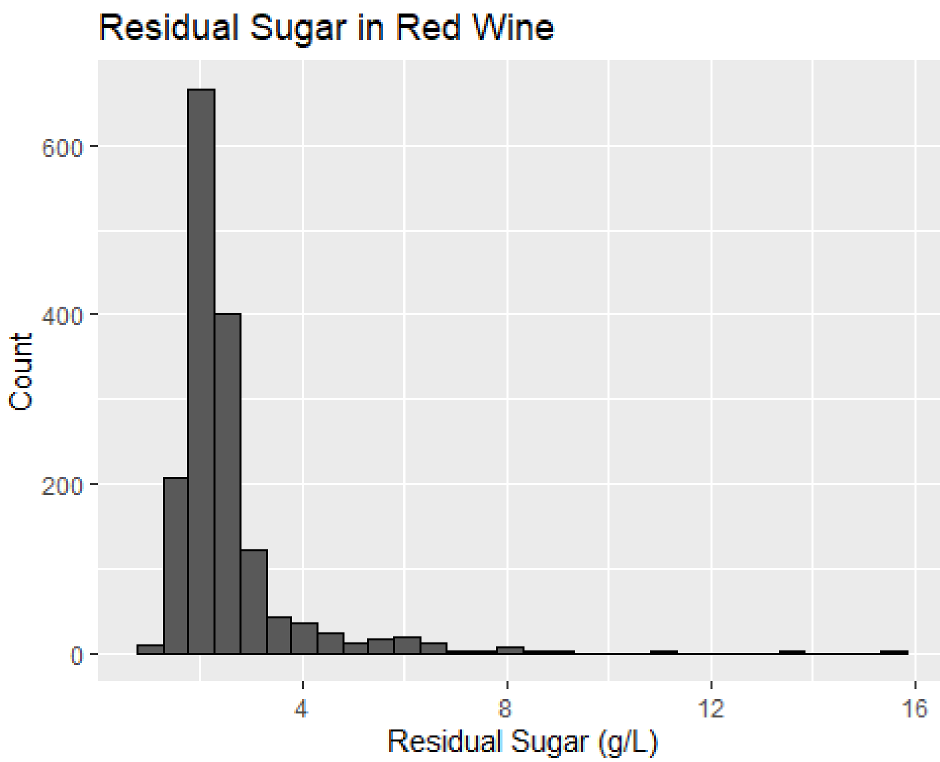

```
ggplot(wine, aes(x = citric.acid)) +
  geom_histogram(color = "black") +
  labs(
    title = "Citric Acid in Red Wine",
    x = "Citric Acid (g/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



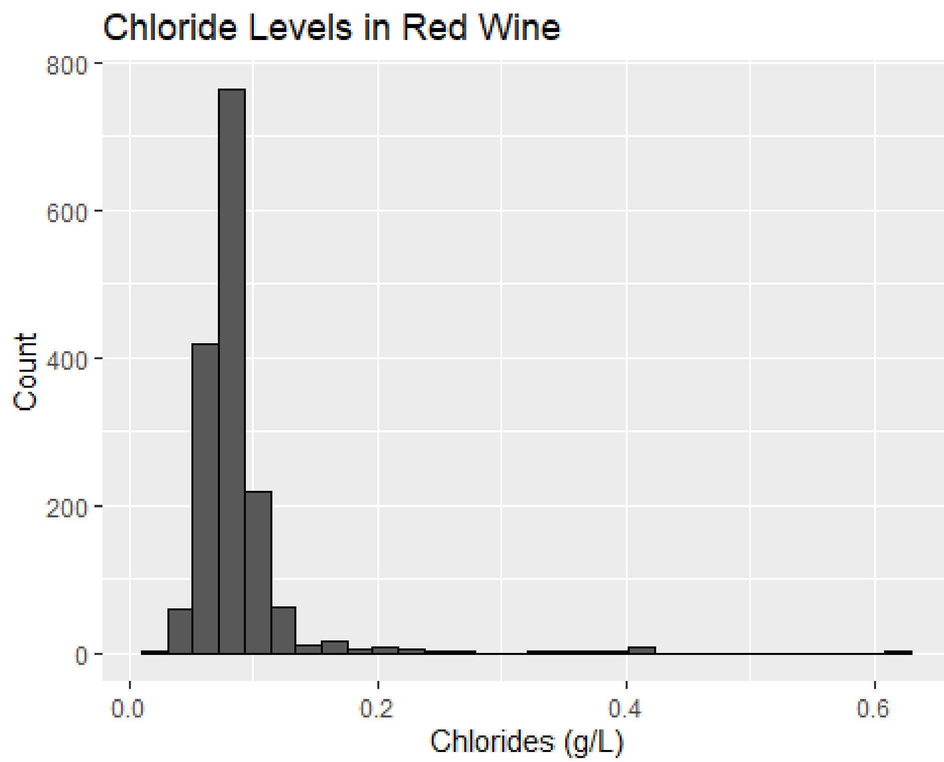

```
ggplot(wine, aes(x = residual.sugar)) +
  geom_histogram(color = "black") +
  labs(
    title = "Residual Sugar in Red Wine",
    x = "Residual Sugar (g/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



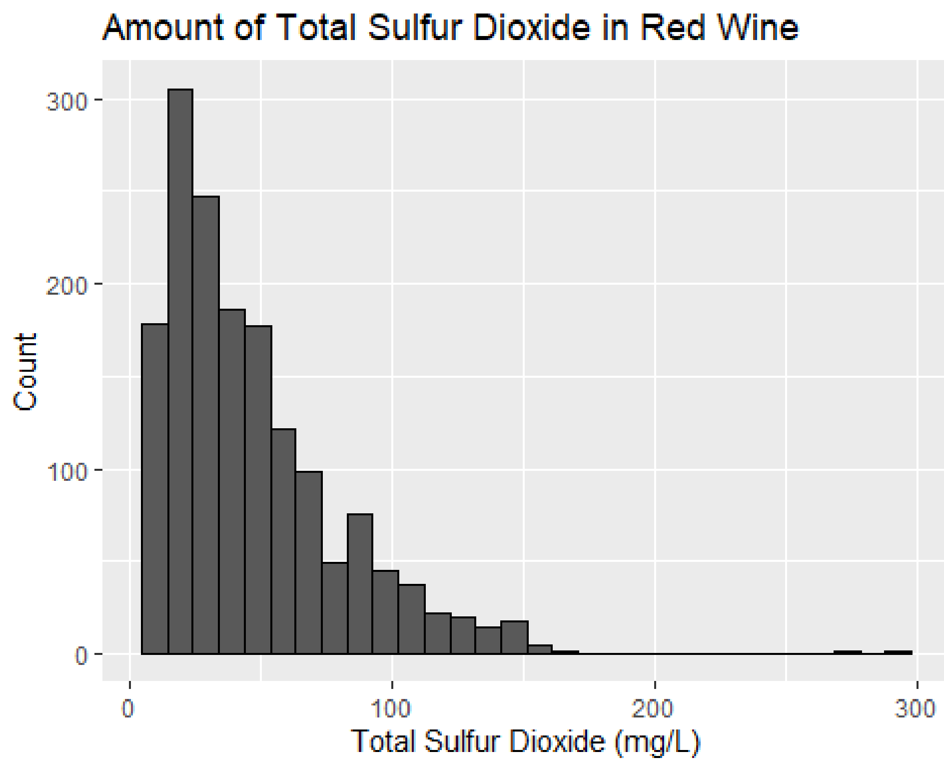

```r
ggplot(wine, aes(x = chlorides)) +
  geom_histogram(color = "black") +
  labs(
    title = "Chloride Levels in Red Wine",
    x = "Chlorides (g/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



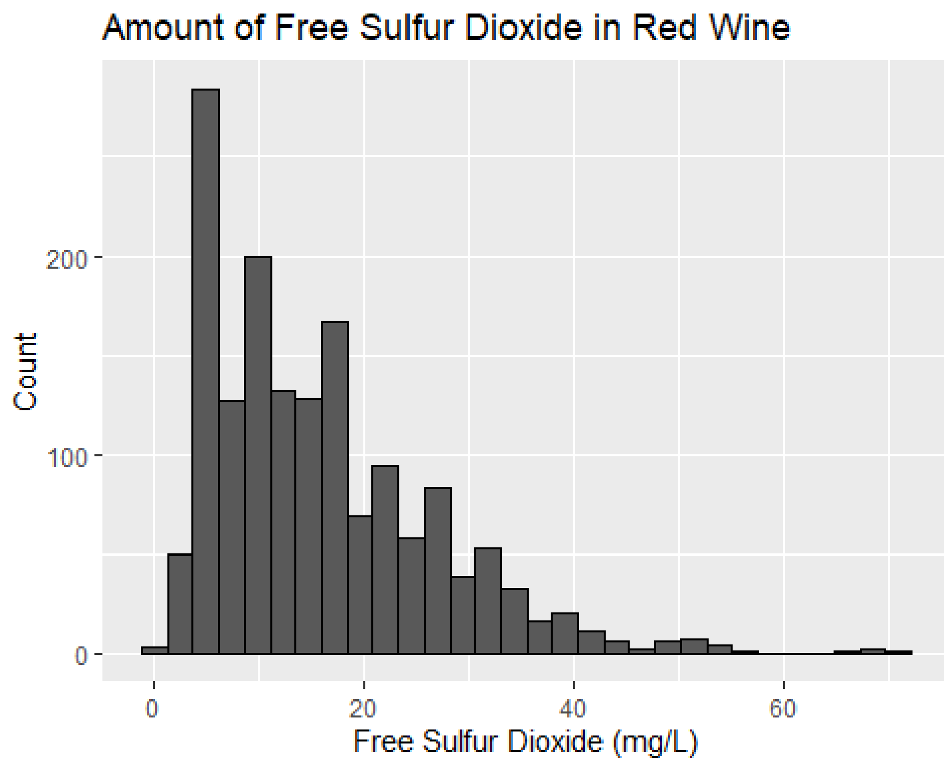

```
ggplot(wine, aes(x = total.sulfur.dioxide)) +
  geom_histogram(color = "black") +
  labs(
    title = "Amount of Total Sulfur Dioxide in Red Wine",
    x = "Total Sulfur Dioxide (mg/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



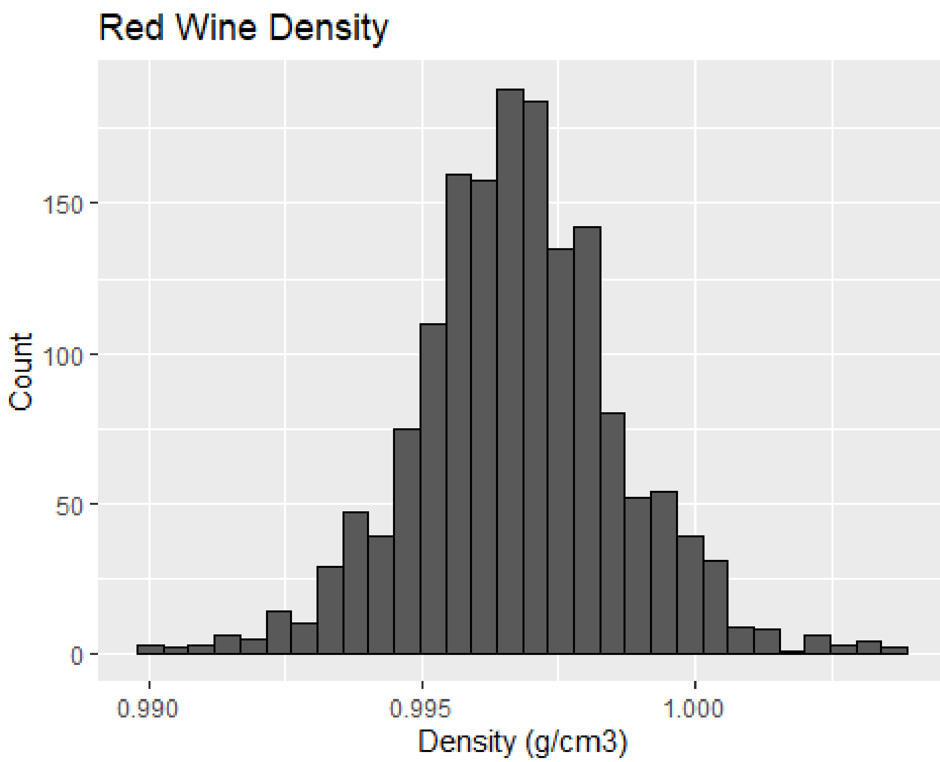

```r
ggplot(wine, aes(x = free.sulfur.dioxide)) +
  geom_histogram(color = "black") +
  labs(
    title = "Amount of Free Sulfur Dioxide in Red Wine",
    x = "Free Sulfur Dioxide (mg/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



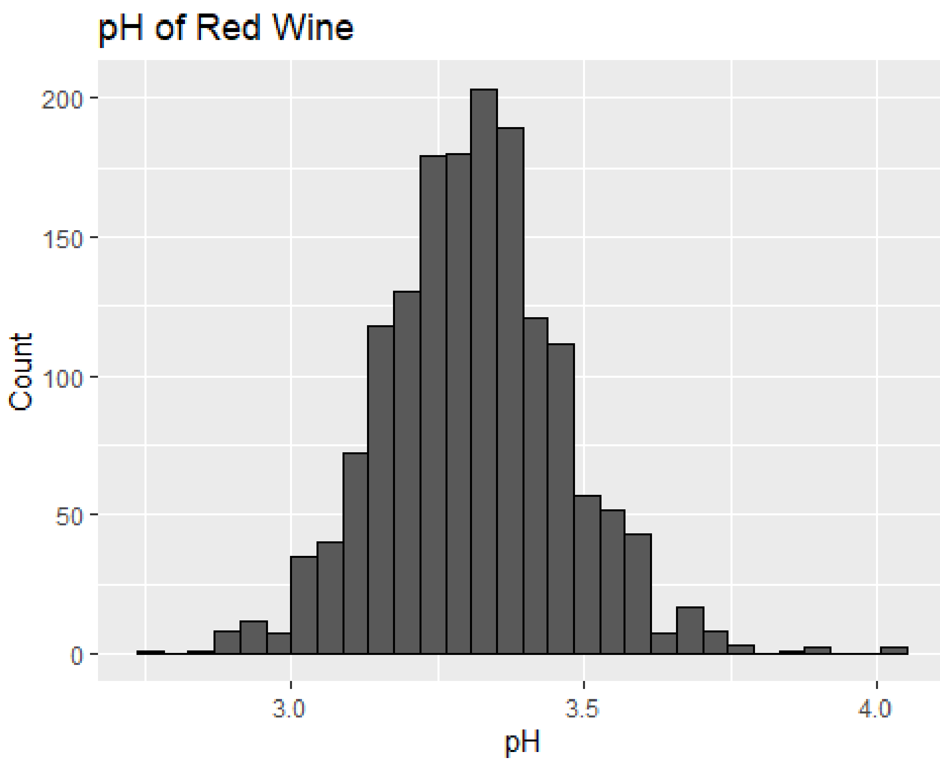

```
ggplot(wine, aes(x = density)) +
  geom_histogram(color = "black") +
  labs(
    title = "Red Wine Density",
    x = "Density (g/cm3)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



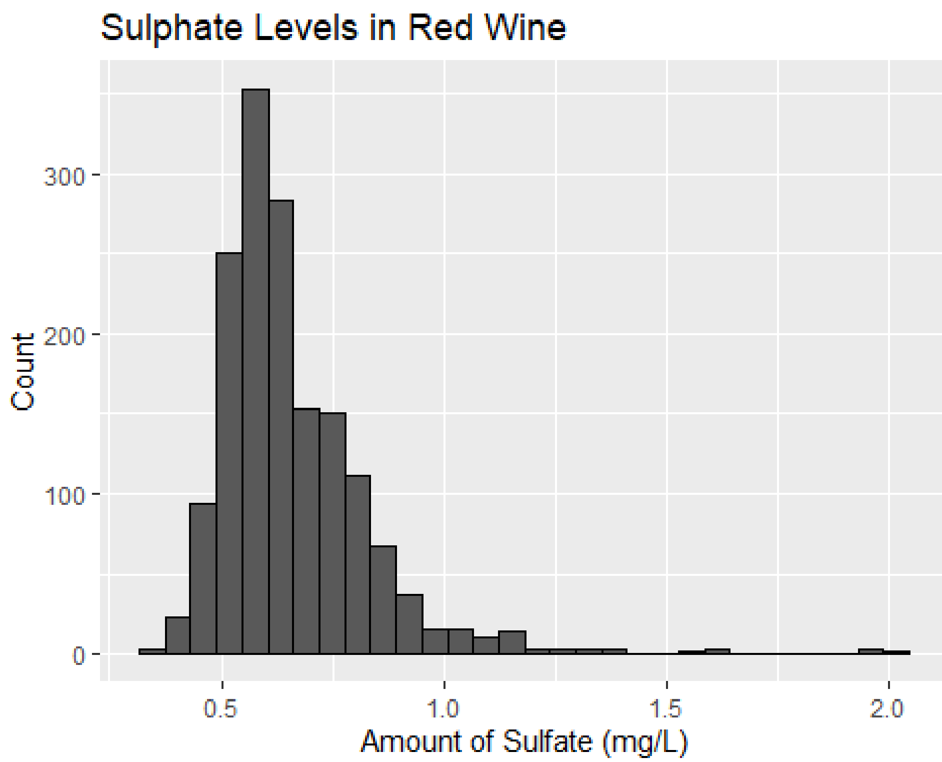

```r
ggplot(wine, aes(x = pH)) +
  geom_histogram(color = "black") +
  labs(
    title = "pH of Red Wine",
    x = "pH",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



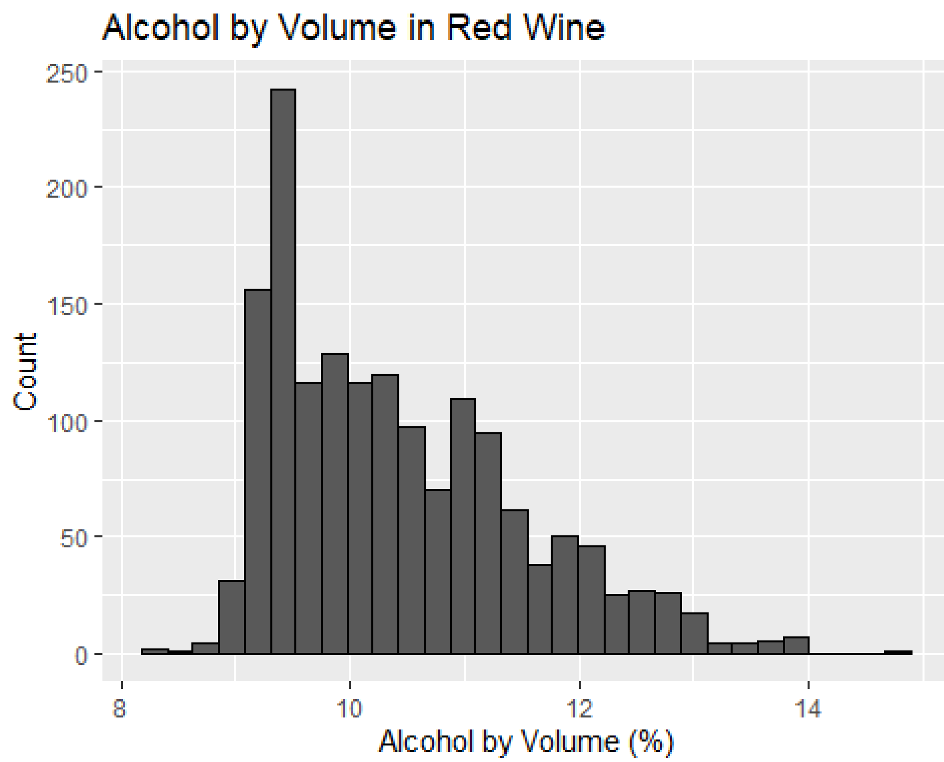

```r
ggplot(wine, aes(x = sulphates)) +
  geom_histogram(color = "black") +
  labs(
    title = "Sulphate Levels in Red Wine",
    x = "Amount of Sulfate (mg/L)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



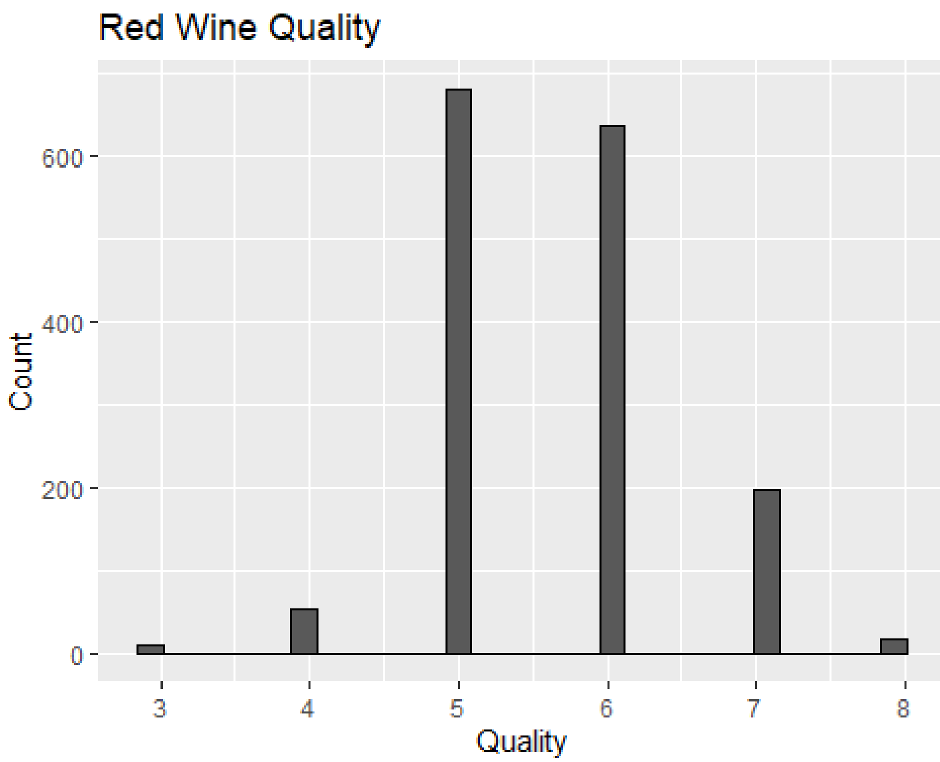

```r
ggplot(wine, aes(x = alcohol)) +
  geom_histogram(color = "black") +
  labs(
    title = "Alcohol by Volume in Red Wine",
    x = "Alcohol by Volume (%)",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
ggplot(wine, aes(x = quality)) +
  geom_histogram(color = "black") +
  labs(
    title = "Red Wine Quality",
    x = "Quality",
    y = "Count"
  )

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



c. Fixed acidity is correlated with citric acid, pH, and density while free and total sulfur dioxide are correlated. Despite these variables being correlated, there are no correlations above or below 0.8 and -0.8, therefore there is no possible multicollinearity indicated.

d. A potential predictor variable that may influence quality is alcohol. It is moderately correlated with quality at 0.47. Looking at the histogram of alcohol, the data is skewed to the right. In this case, looking at the outlier numbers, they aren't out of the scope of alcohol percentage in wines, therefore they don't need to be ignored or removed. Using a log scale could reduce skeweness for skewed data but in this case, it doesn't make sense since alcohol's median is 10.20 and mean is 10.42, which are quite similar therefore shouldn't majorly affect modeling.

Fit the Full Regression Model

```
model_full <- lm(quality ~ ., data = wine)
summary(model_full)
```

```
##
## Call:
## lm(formula = quality ~ ., data = wine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.68911 -0.36652 -0.04699  0.45202  2.02498
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          2.197e+01  2.119e+01   1.036   0.3002
## fixed.acidity        2.499e-02  2.595e-02   0.963   0.3357
## volatile.acidity    -1.084e+00  1.211e-01  -8.948  < 2e-16 ***
## citric.acid         -1.826e-01  1.472e-01  -1.240   0.2150
## residual.sugar       1.633e-02  1.500e-02   1.089   0.2765
## chlorides           -1.874e+00  4.193e-01  -4.470 8.37e-06 ***
## free.sulfur.dioxide  4.361e-03  2.171e-03   2.009   0.0447 *
## total.sulfur.dioxide -3.265e-03  7.287e-04  -4.480 8.00e-06 ***
## density             -1.788e+01  2.163e+01  -0.827   0.4086
## pH                  -4.137e-01  1.916e-01  -2.159   0.0310 *
## sulphates            9.163e-01  1.143e-01   8.014 2.13e-15 ***
## alcohol              2.762e-01  2.648e-02  10.429  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.648 on 1587 degrees of freedom
## Multiple R-squared:  0.3606, Adjusted R-squared:  0.3561
## F-statistic: 81.35 on 11 and 1587 DF,  p-value: < 2.2e-16
```

b. Based on the full regression model, p-values with negative coefficients can significantly influence wine quality negatively if they are less than 0.05. Similarly, p-values with positive coefficients that are less than 0.05 will have a positive effect on wine quality. Predictors such as volatile acidity, chlorides, total sulfur dioxide, and pH negatively affect wine quality. Predictors such as free sulfur dioxide, sulphates, and alcohol positively affect wine quality. Predictors such as fixed acidity, citric acid, residual sugar, and density do not significantly affect wine quality.

Model Selection

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
model_best <- stepAIC(model_full, direction = "both", trace = FALSE)
summary(model_best)

##
## Call:
## lm(formula = quality ~ volatile.acidity + chlorides + free.sulfur.dioxide
+
##     total.sulfur.dioxide + pH + sulphates + alcohol, data = wine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.68918 -0.36757 -0.04653  0.46081  2.02954
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          4.4300987  0.4029168  10.995  < 2e-16 ***
## volatile.acidity    -1.0127527  0.1008429 -10.043  < 2e-16 ***
## chlorides           -2.0178138  0.3975417  -5.076 4.31e-07 ***
## free.sulfur.dioxide  0.0050774  0.0021255   2.389    0.017 *
## total.sulfur.dioxide -0.0034822  0.0006868  -5.070 4.43e-07 ***
## pH                  -0.4826614  0.1175581  -4.106 4.23e-05 ***
## sulphates            0.8826651  0.1099084   8.031 1.86e-15 ***
## alcohol              0.2893028  0.0167958  17.225  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6477 on 1591 degrees of freedom
## Multiple R-squared:  0.3595, Adjusted R-squared:  0.3567
## F-statistic: 127.6 on 7 and 1591 DF,  p-value: < 2.2e-16
```

a. Stepwise selection creates a regression model by either starting with just the y-intercept then adding variables to the model that improve the model, and by starting with all variables and removing the ones that do not improve the equation until the model can most accurately predict the system. StepwiseAIC uses the AIC value (Akaike information criterion) to determine if the variable has improved the model or not, with AIC being a measure of both goodness of fit and parsimonious-ness.
b. quality = 4.43 - 1.013(x1) - 2.018(x2) + 0.005(x3) - 0.003(x4) - 0.483(x5) + 0.883(x6) + 0.289(x7)

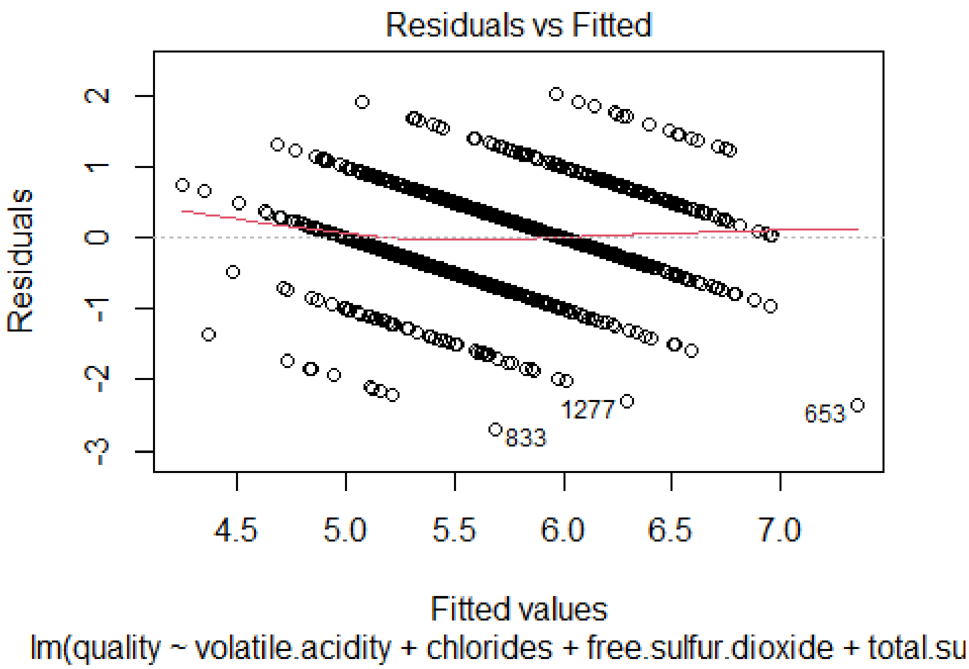
$$quality = 4.43 - 1.013x_1 - 2.018x_2 + 0.005x_3 - 0.003x_4 - 0.483x_5 + 0.883x_6 + 0.289x_7$$

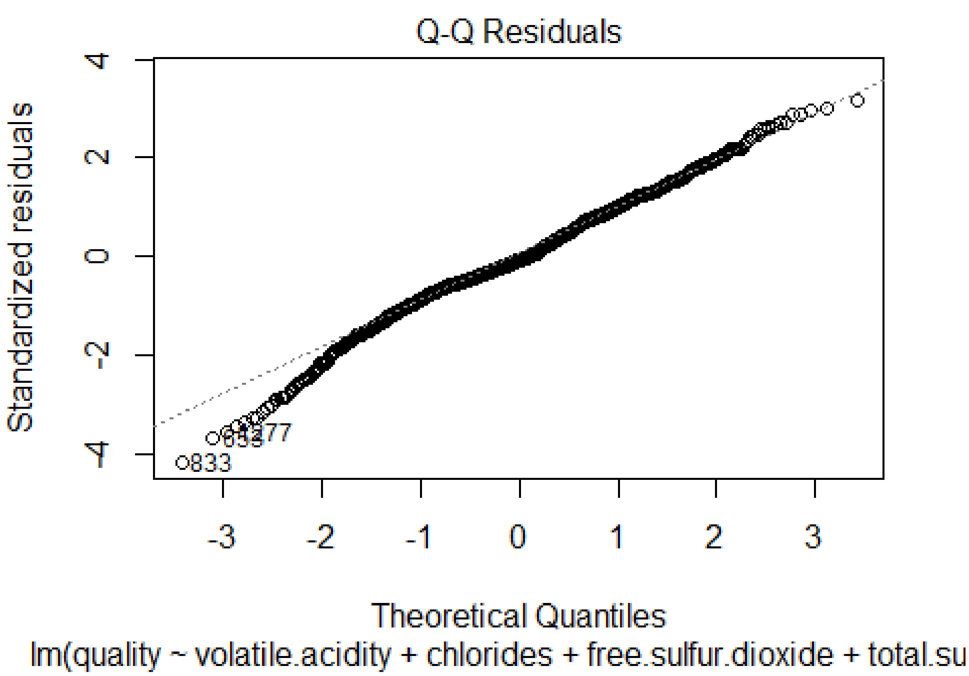
Where $x_1 = \text{volatile acidity};\ x_2 = \text{chlorides};\ x_3 = \text{free sulfur dioxide};\ x_4 = \text{total sulfur dioxide};\ x_5 = \text{pH};\ x_6 = \text{sulphates};\ x_7 = \text{alcohol}$

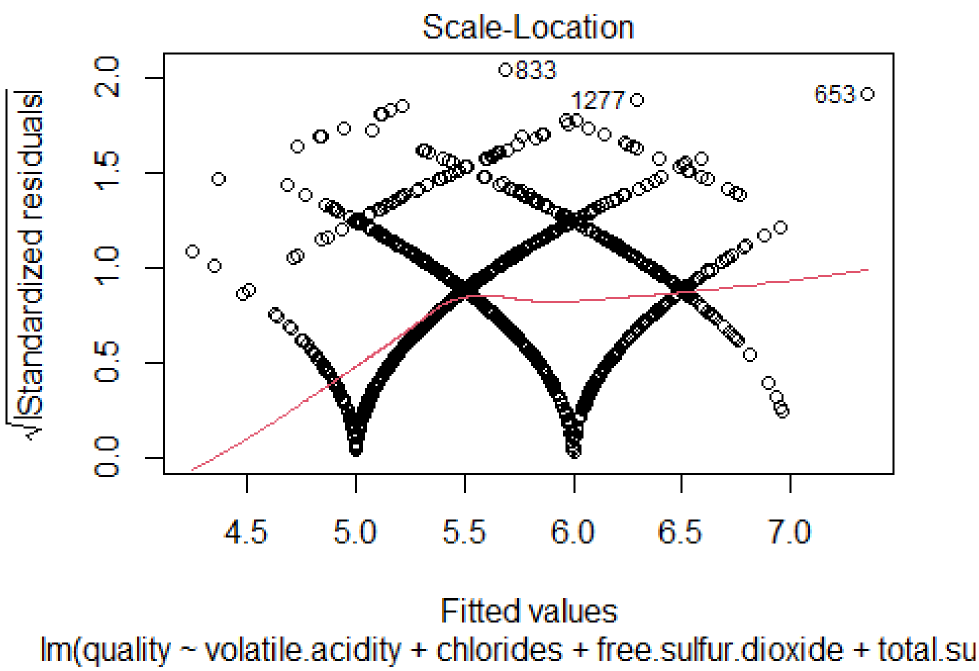
c. The adjusted R-squared value for model_full is 0.3561 and the value for model_best is 0.3567. Therefore, model_best is a better model for predicting wine quality. However, the differences in values is very small, so model_best is not much better, but it more parsimonious as it has 8 predictors instead of 11.

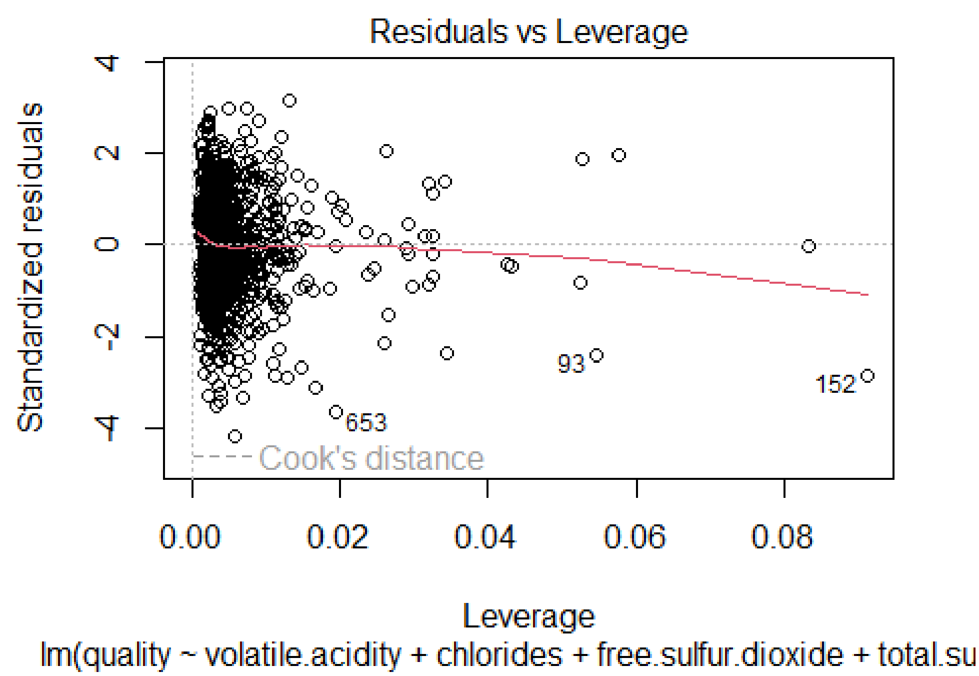
Model Diagnostics

```
plot(model_best)
```


Residuals vs Fitted
Residuals
Fitted values
lm(quality ~ volatile.acidity + chlorides + free.sulfur.dioxide + total.su
1277
833
653

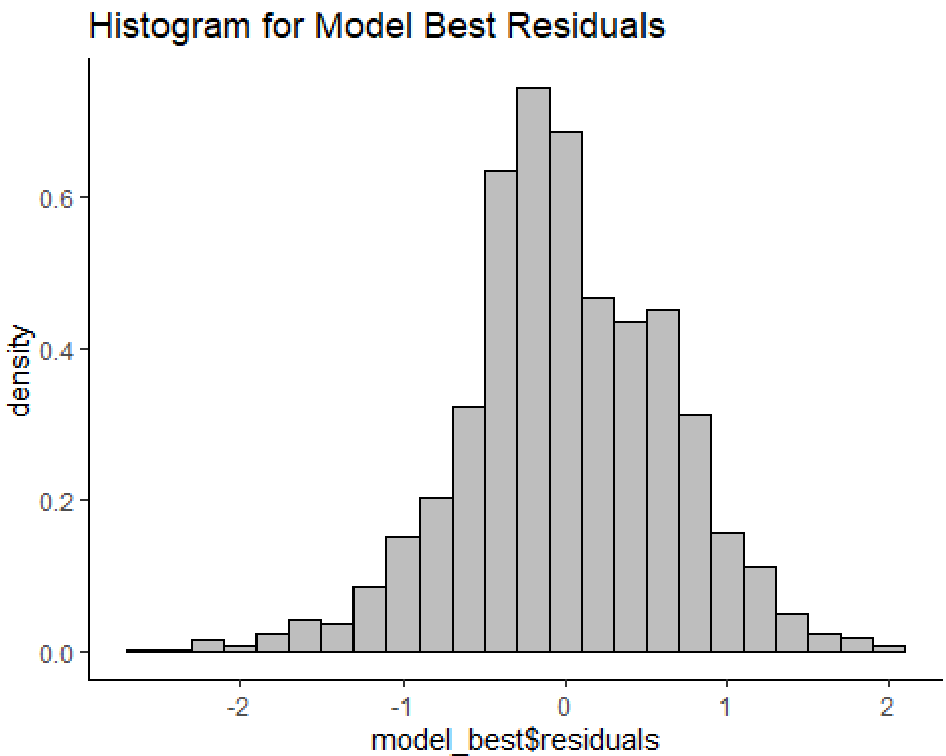

Q-Q Residuals
Standardized residuals
Theoretical Quantiles
lm(quality ~ volatile.acidity + chlorides + free.sulfur.dioxide + total.su
833






```
library(ggplot2)
ggplot(data = wine, aes(model_best$residuals))+
  geom_histogram(aes(y=after_stat(density)), binwidth = 0.2, color = 'black',
fill = 'grey')+
```

```
  theme(panel.background = element_rect(fill = 'white'),
        axis.line.x=element_line(),
        axis.line.y=element_line()) +
  ggtitle("Histogram for Model Best Residuals")
```



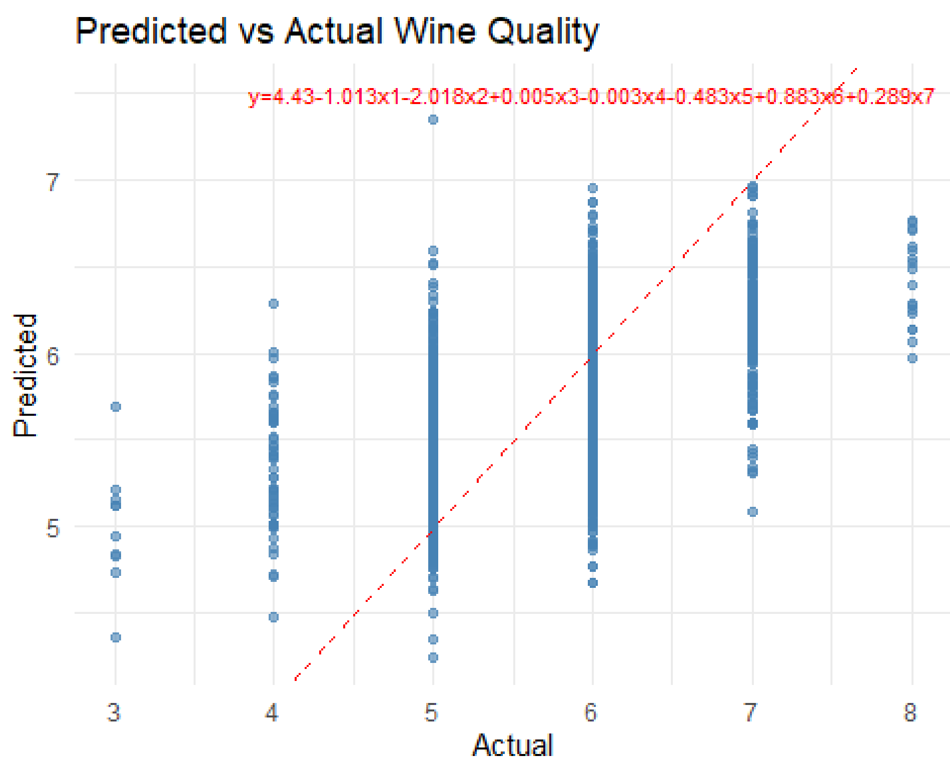
c) It appears as though these data have met the assumptions for regression analysis. Although the residual vs. fitted plot looks like it shows a pattern, it only doing so because quality is a limited variable, so it does have homoscedasticity. Furthermore, these data are fairly normal – as seen in both the Q-Q plot and the histogram – and there is no multicollinearity – as we discussed earlier. The model is linear and the data is independent.

d) The residuals vs. fitted plot is unusual. It shows residuals plotted in 6 neat diagonal lines above and below 0. Typically, Residuals vs Fitted plots show residuals places randomly around 0 to indicate that the residuals are normally distributed. In our data, however, wine quality is quantified by 6 integers, meaning the residuals have to be lined up instead of scattered around the plot. If the quality was measured with a more detailed number, we would likely not see the 'pattern' we see in our plot.

Visualization

```
wine$predicted <- predict(model_best)
ggplot(wine, aes(x = quality, y = predicted)) +
```

```r
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  labs(title = "Predicted vs Actual Wine Quality",
       x = "Actual", y = "Predicted") +
  annotate("text", x = 6, y = 7.5, label = "y=4.43-1.013x1-2.018x2+0.005x3-0.
003x4-0.483x5+0.883x6+0.289x7", color = "red", size = 3) +
  theme_minimal()
```



b. The plot above shows the regression model of the data. While the model performs okay, it could be better. The model shows a positive linear regression, accurately showing the prediction that higher quality wines are actually higher quality. There are several data points close to the line for a 1:1 regression, indicating that the model predicts quality well. However, there are many points not close to the line and scattered in vertical lines, which means that the model makes errors when predicting the quality of wine. This shows that wine quality is affected by other predictors not used in the data set.

Interpretation and summary

According to the data, the chemical features that significantly predict red wine quality are: volatile acidity, chlorides, total sulfur dioxide, pH, free sulfur dioxide, sulphates, and alcohol. By looking at the equation, we can say that higher free sulfur dioxides, sulfates, and alcohol would predict a better quality wine, and higher volatile acidity, total sulfur dioxides, chlorides, and pH would predict a lower quality wine. Chlorides and volatile acidity most negatively affect wine quality, and alcohol and sulphates most positively affect wine quality, which is consistent with the data.

Although our model meets the assumptions of regression analysis and has an adjusted R-squared value of 0.3567, this model is not the best predictor for the quality of red wine. The model shows a trend in the data, but it also has unexplained variation shown by the scattered data points in the regression model. One reason for this is the limited and subjective scale of ‘quality’ given by the data set. If there were a more nuanced or less subjective measurement of red wine quality, the model’s predictive value would likely improve.